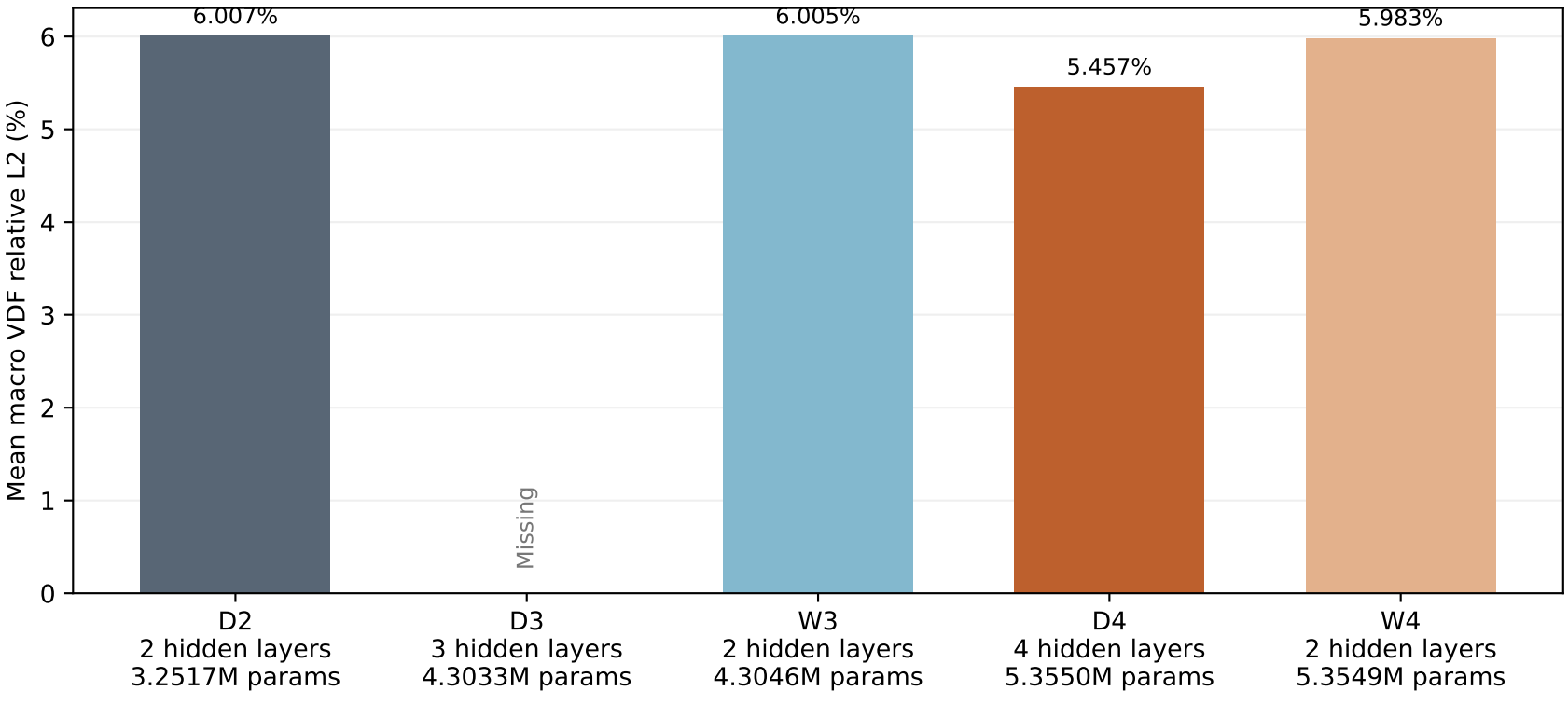

Recovery: 8 byte-identical historical checkpoints; 3 final checkpoints unavailable. Some training tails are incomplete.
Existing reused test n=1024; Fourier-label VDF errors; lower is better. Historical D4 repeat family retained.
Macro: equal mean of electron, H and C errors per sample; no error bars.